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Home Assignment :- 4

Q A] Examine whether following mapping is L.T. or not.

4) $T: R^4 \to R^3$ Defined as $T\left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}\right) = \left(\begin{bmatrix} x \\ x+z \\ y+w \\ 0 \end{bmatrix}\right)$

→ $T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$ — Additivity.

$T(c\vec{u}) = cT(\vec{u})$ —— Homogeneity.

Let

$\vec{u} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ w_1 \end{bmatrix}$ & $\vec{v} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ w_2 \end{bmatrix}$

$$T(\vec{u}+\vec{v}) = T\left(\begin{bmatrix} x_1+x_2 \\ y_1+y_2 \\ z_1+z_2 \\ w_1+w_2 \end{bmatrix}\right) = \begin{bmatrix} x_1+x_2 \\ (x_1+x_2)+(z_1+z_2) \\ (y_1+y_2)+(w_1+w_2) \end{bmatrix}$$

$$= \begin{bmatrix} x_1 \\ x_1+z_1 \\ y_1+w_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ x_2+z_2 \\ y_2+w_2 \end{bmatrix} = \underline{T(\vec{u})+T(\vec{v})}$$

$$T(c\vec{u}) = T\left(\begin{bmatrix} cx_1 \\ cy_1 \\ cz_1 \\ cw_1 \end{bmatrix}\right) = \begin{bmatrix} cx_1 \\ cx_1+cz_1 \\ cy_1+cw_1 \end{bmatrix} = c\begin{bmatrix} x_1 \\ x_1+z_1 \\ y_1+w_1 \end{bmatrix}$$

$$= \underline{cT(\vec{u})}$$

∴ Transformation T is a Linear Transformatⁿ as it satisfies both additivity & homogeneity.

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Q B] Consider L.T. $T(x) = Ax$

Find i) Basis of $\ker(T)$ iii) Rank(T)

ii) nullity of T iv) verify Rank Nullity Th^m

$\longrightarrow$

3 a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$ This 3×2 matrix so,

$T : R^2 \to R^3$

i) Basis of ker (T) ⇒

$Ax = 0$, where $x = \begin{bmatrix} x \\ y \end{bmatrix}$

$\therefore \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ x+y \\ x+y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow x+y=0$

$\therefore \ker(T) = \left\{ \begin{bmatrix} x \\ -x \end{bmatrix} : x \in R \right\} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

$\therefore$ Basis of $\ker(T) \Rightarrow \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

ii) Nullity of T is dimension of kernel.

$\therefore$ nullity $(T) = 1$.

iii) Rank of T.

rank (T) + nullity (T) = dimension of domain $= 2$

$\Rightarrow$ rank$(T) = 2 - 1 \Rightarrow 1$.

$\therefore$ rank$(T) = 1$.

iv) Rank - Nullity Theorem.

rank(T) + nullity(T) ⇒ 1 + 1 ⇒ 2 = dim(R^2)

Q c] Geometric L.T.

Find matrix of L.T. $T: R^3 \to R^2$ which produces effect of shear of factor 2 along +ve x directⁿ, followed by reflectⁿ about $y=-x$, followed by clockwise rotatⁿ through an angle of 60°. Hence find image of $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ under transformatⁿ T.

→

- shear in x directⁿ with factor 2.
- Reflectⁿ about line $y = -x$.
- Clockwise rotatⁿ by 60°.

⇒

$u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\quad \therefore S = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

Matrix for reflectⁿ about $y = -x$ ⇒

$R = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

Matrix for 60° clockwise rotatⁿ ⇒

$\theta = \frac{\pi}{3}$, $\cos\theta = \frac{1}{2}$ & $\sin\theta = \frac{\sqrt{3}}{2}$

$\therefore C = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ $\quad \therefore C = \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}$

∴ Transformatⁿ matrix, $T = C.R.S.$

$R.S. \Rightarrow \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -1 \\ -1 & -2 \end{bmatrix}$

$\therefore C.R.S. \Rightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -1 & -2 \end{bmatrix} \Rightarrow \begin{bmatrix} -\sqrt{3}/2 & -1/2 - \sqrt{3} \\ -1/2 & \sqrt{3}/2 - 1 \end{bmatrix}$

Image of $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$\therefore \; Tu = \begin{bmatrix} -\sqrt{3}/2 & -1/2 - \sqrt{3} \\ -1/2 & \sqrt{3}/2 - 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 - 5\sqrt{3}/2 \\ \sqrt{3} - 5/2 \end{bmatrix}$$